

Shell-and-Tube Heat Exchanger Performance

Using Aspen Plus — HEATX Block (Water → Benzene)

Stage-2

Group Members:

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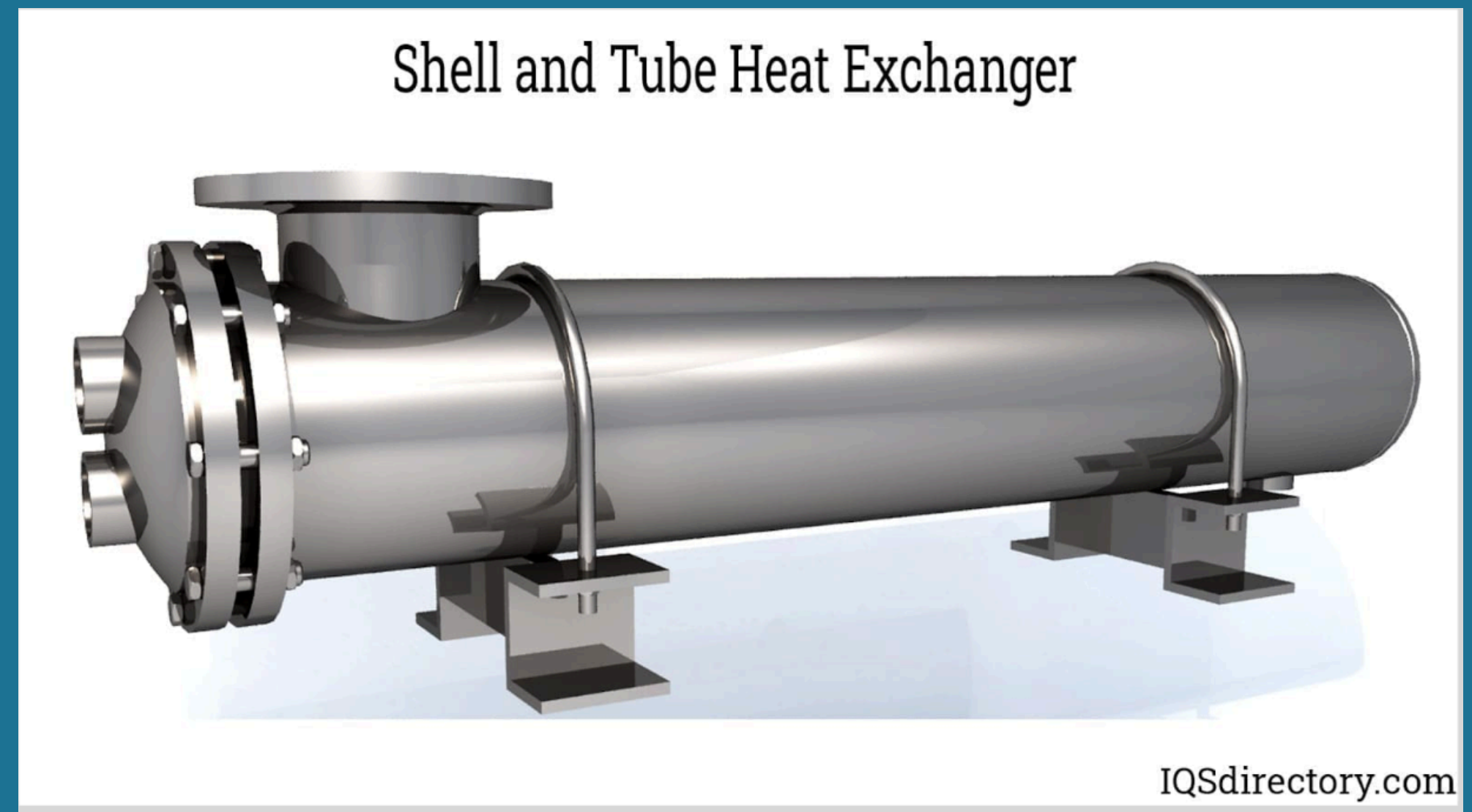
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Objective & Problem Statement

Primary Objective

To study the effect of cooling water flow rate on the overall heat transfer coefficient (**U**) in a shell-and-tube heat exchanger using Aspen process simulation software.

The cooling water flow rate is considered as the independent variable in this study.

Core Hypothesis

Increasing the cooling water flow rate increases fluid velocity, leading to higher turbulence and Reynolds number, which enhances heat transfer coefficient.

□ **Key relationship:** Flow Rate $\uparrow \rightarrow$ Velocity $\uparrow \rightarrow$
Turbulence $\uparrow \rightarrow U \uparrow$
i.e. $U \propto \text{Flow Rate}$

Motivation & Real-World Context

Why heat exchangers matter in industry



Process Industry Applications

- Widely used in refineries, chemical plants, and power plants
- Essential for heating, cooling, and condensation operations



Energy Efficiency

- Reduces energy consumption
- Lowers operating cost
- Minimizes carbon footprint



Design & Performance Control

- Flowrate controls Reynolds number and turbulence
- Directly influences heat transfer coefficient (U)
- Critical for exchanger performance

This Study in Industrial Context

- Represents a real refinery/chemical plant operation: cooling of a hydrocarbon stream (benzene)
- Hot utility (steam/water) transfers heat to a cold process stream in a counter-current shell-and-tube exchanger
- Flow rate is a key operating variable used in industry to control heat transfer performance
- This study quantifies how variation in flow rate affects U , which is critical for exchanger design, rating, and optimization
- Results can be directly applied in industrial sizing, performance analysis, and energy optimization

Relevant Background & Theory

Heat exchanger fundamentals and literature

Governing Equations

$$Q = U \cdot A \cdot \Delta T_{\text{LMTD}}$$

Fundamental heat exchanger equation

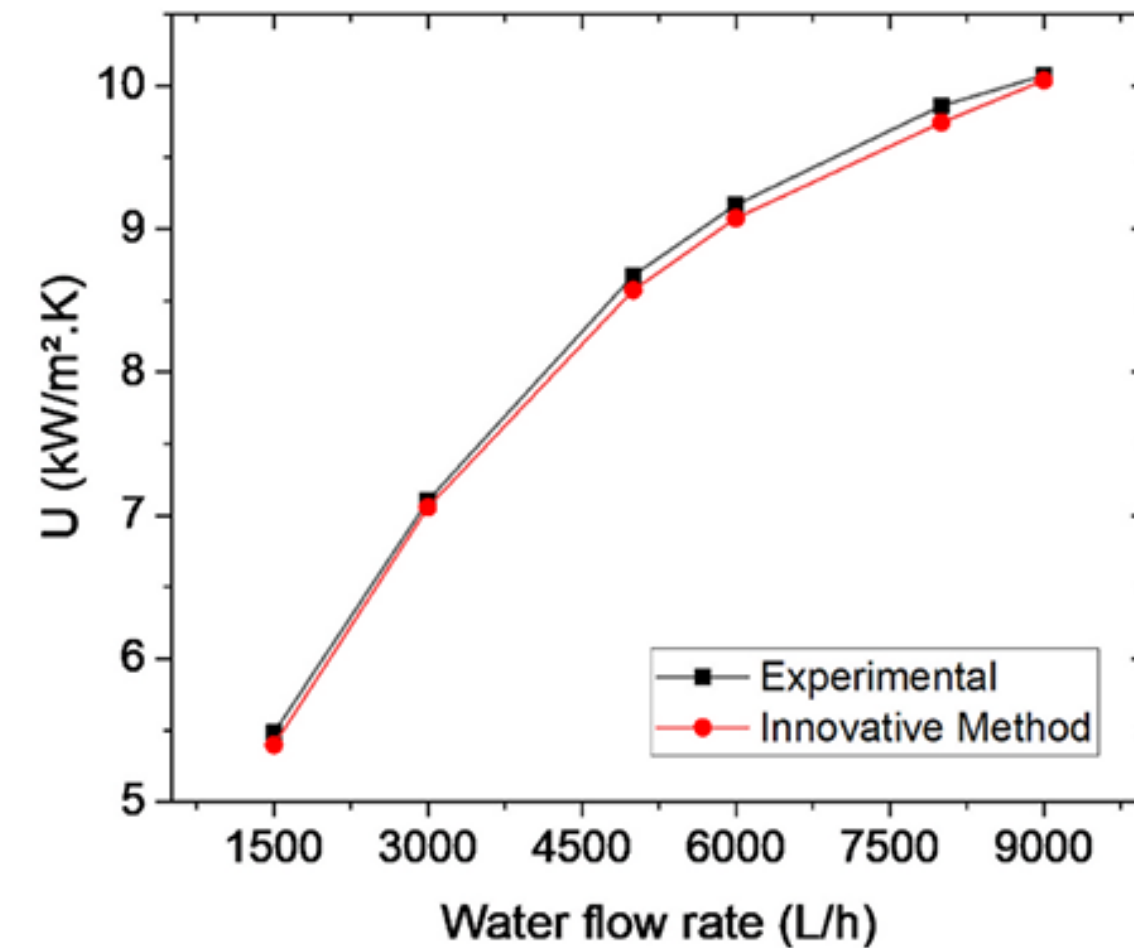
$$\Delta T_{\text{LMTD}} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$$

Log mean temperature difference

$$Q = \dot{m} \cdot C_p \cdot \Delta T$$

Sensible heat duty for each stream

These equations are used to verify Aspen Plus results through theoretical validation (LMTD and energy balance)



Literature & References

[1] Faraj, A., et al. (2021). *Case Studies in Thermal Engineering*, 28, 101434.

- Validated U prediction using LMTD method (error < 1.6%)
- Our simulation trend aligns with published experimental trend, validating our model

Objectives & Project Overview

01 Model Setup

- Develop a counter-current shell-and-tube heat exchanger model in Aspen Plus
- Fluids used: water (cooling) and benzene (process stream)

02 Thermal Analysis

Evaluate key parameters: heat duty (Q), LMTD, and overall heat transfer coefficient (U).

03 Geometry & EDR

Investigate the effect of cooling water flow rate on heat transfer coefficient (U) and exchanger performance.

04 Validation

Validate simulation results against theoretical hand calculations using standard heat transfer equations.

05 Sensitivity

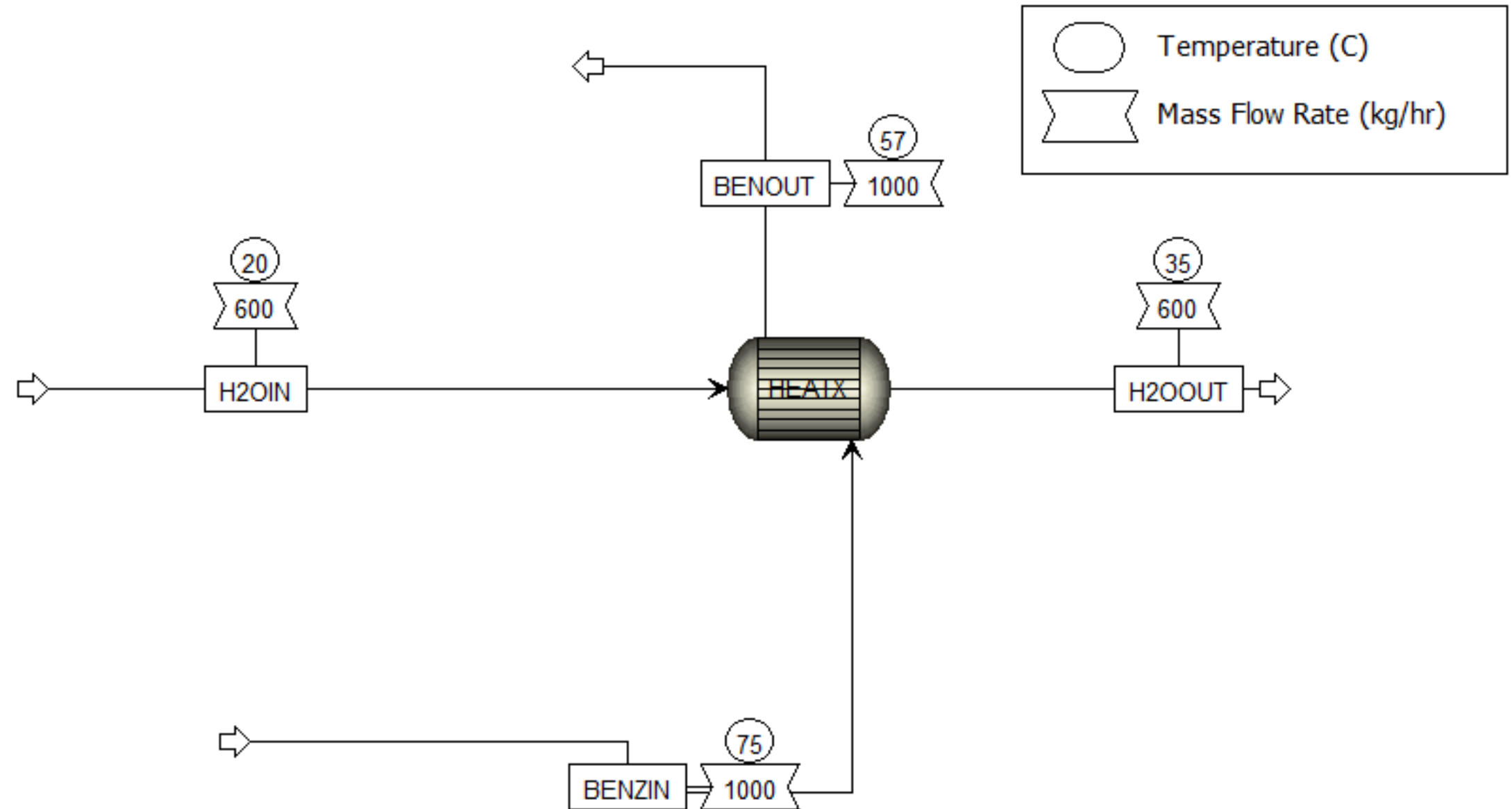
Validate simulation results using LMTD calculation and energy balance ($Q = \dot{m}C_p\Delta T$).

ASPEN FLOWSHEET

HEATX flowsheet

Process Description:

- Hot process stream: Benzene (inlet ~75°C)
- Cooling medium: Water (inlet ~20°C)
- Heat exchanger: Shell-and-tube (counter-current)
- Objective: Cool benzene using water



Aspen Plus flowsheet of shell-and-tube heat exchanger used for cooling benzene with water (counter-current configuration).

Relevant Model Details

Stream specifications and HEATX block configuration

Parameter	Cold (H2OIN)	Hot (BENZIN)
Component	Water	Benzene
Inlet Temp	20C	75C
Pressure	3 bar	2.5 bar
Molar Flow	33.3 kmol/hr	12.8 kmol/hr
Liquid Frac.	1	1
Outlet Temp	35 C	57 C

◆ HEATX Configuration

- Model: Aspen Plus HeatX
- Flow: Counter-current
- Property Method: NRTL
- Heat Loss: Adiabatic

◆ Key Performance Parameter

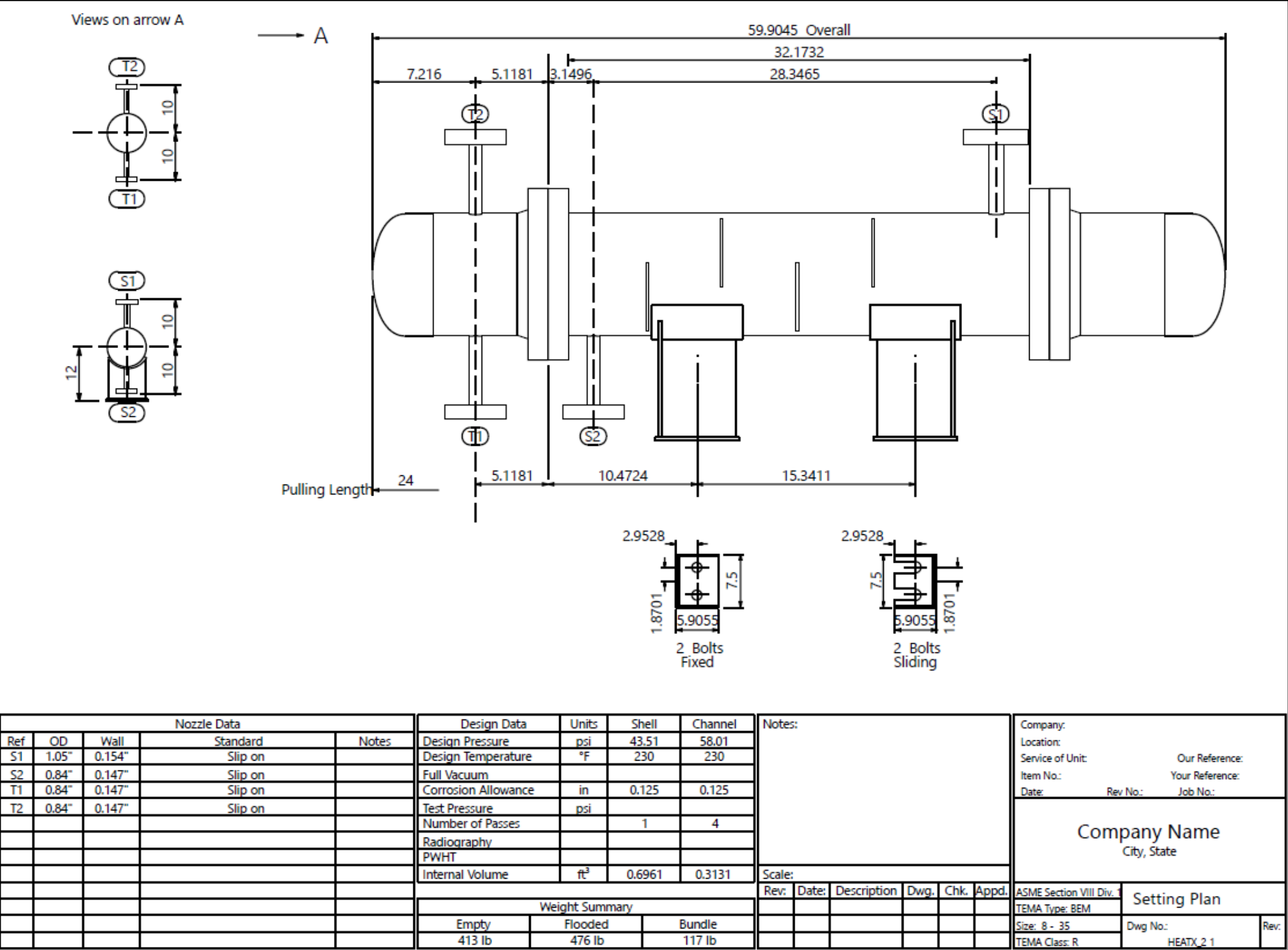
- Heat Duty: 0.0086 Gcal/hr
- U: 225.79 kcal/hr·m²·K
- LMTD: 37.99°C
- Area: ~1 m²

◆ Geometry

- TEMA Type: BEM
- Hot fluid: Shell side
- Tube OD / Pitch: 0.75 / 0.9375 in
- Tubes: 20
- Tube passes: 4
- Baffles: 6

EDR Geometry & Shell Design

Aspen EDR shell-and-tube sizing results

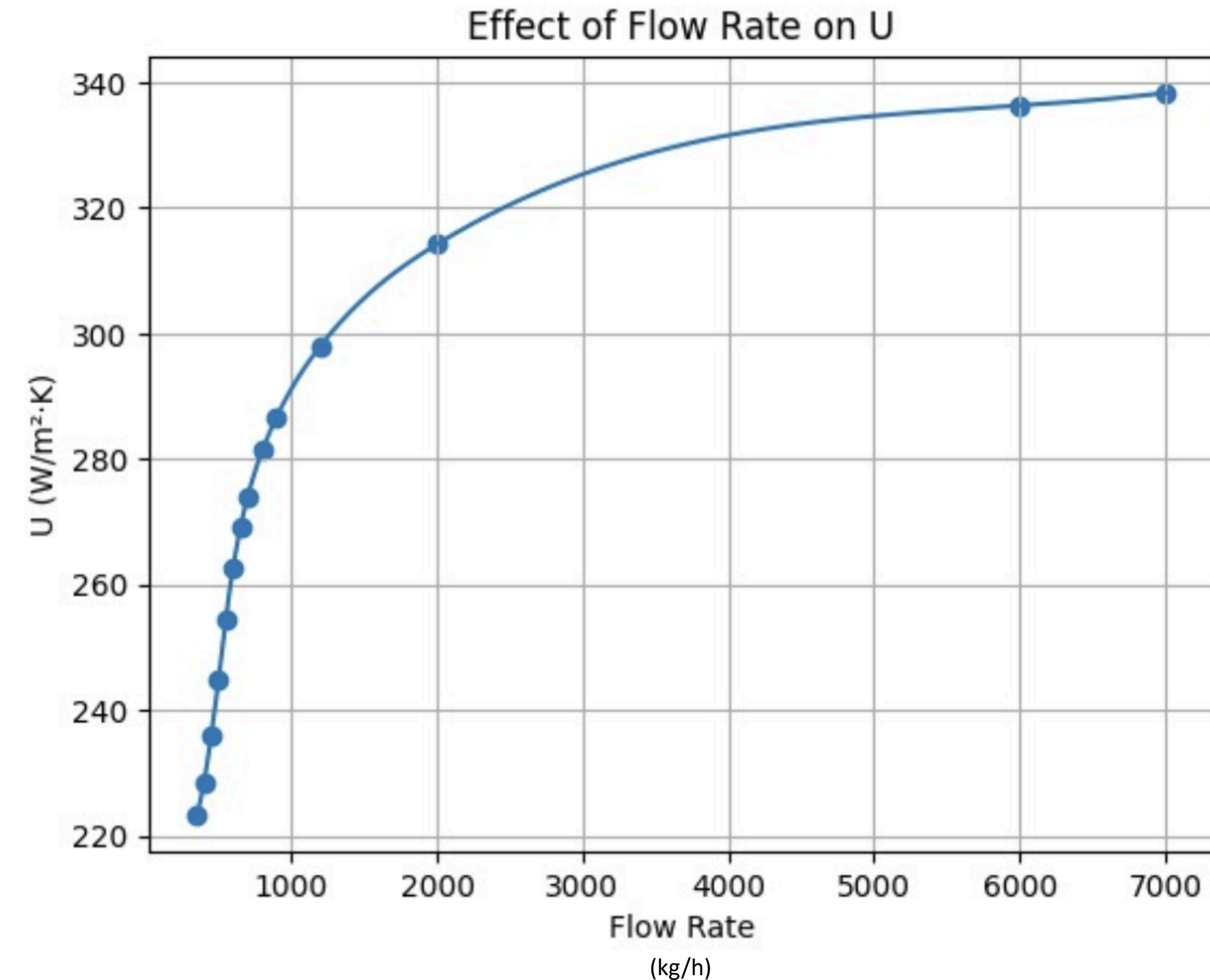


Aspen EDR schematic of shell-and-tube heat exchanger showing optimized geometry and shell-side configuration (TEMA BEM)

Key EDR Results

Parameter	Value
Shell ID / OD	8.071 / 8.625 in
Tube Length	35.244 in
Tube Count / Passes	20 tubes, 4 passes
Baffle Spacing	5.315 in
Number of Baffles	6
Excess Surface	-2 %
Total Cost	\$8,766 USD
ΔP Ratio (Shell / Tube)	0.0823 / 0.0248
Heat Transfer Area: ²	~1.0 m ²

Effect of Flow Rate on Overall Heat Transfer Coefficient (U)



Key Observations

- As flow rate increases, U increases significantly at first.
- After a certain point, the increase in U becomes gradual (plateau region).
- At high flow rates, further increase has minimal effect on U.

Reason for Trend

- Higher flow rate → more turbulence → better heat transfer → higher U
- At very high flow rates:
- Flow is already turbulent
- Heat transfer resistance becomes limited by other factors
- Hence, diminishing returns

Theoretical Validation

Hand calculations vs. Aspen Plus simulation results

Simulation vs. Theory Comparison

Metric	Hand Calc.	Aspen Plus	Match?
LMTD (°C)	38.5	37.99	✓ Yes
Heat Duty	9.0 kW	8.60 kW	✓ Yes
U (W/m²·K)	~220	225.79	✓ Yes
F-factor	≈ 1.0	0.999	✓ Yes
Req. Area (m²)	~1.002	0.979	✓ Yes

Conclusion: Aspen Plus results align well with hand calculations. Minor discrepancies in required area stem from differences in assumed U-value and Cp. The simulation is validated as accurate for this thermal design.

Individual Contributions

Group member responsibilities and deliverables

Abhinav	Kushagra & Aarav	Soham	Sohel	Prakrut
<i>Simulation Setup & Flowsheet</i>	<i>EDR Sizing & Geometry</i>	<i>Results Analysis</i>	<i>Validation & Hand Calculations</i>	<i>Presentation & Literature</i>
→ Abhinav developed the Aspen Plus flowsheet, defined stream specifications, configured the HeatX block, and performed the initial simulation.	→ Kushagra and Aarav set up the Aspen EDR exchanger, configured the TEMA BEM geometry, analyzed tube layout and baffles, and reported the geometry results.	→ Soham performed sensitivity analysis of flowrate on the overall heat transfer coefficient (U), generated U vs. flowrate plots, analyzed simulation scenarios, and identified over-design issues.	→ Soheli carried out hand calculations, validated LMTD, heat duty, and U with theoretical values, cross-checked simulation outputs, and documented the validation process.	→ Prakrut prepared the presentation slides, compiled literature references, formatted the report, and coordinated group submissions.

[5] Additional Research Papers

- Fares, M., et al. (2020). Experimental investigation of flow rate effect on heat transfer in shell-and-tube heat exchangers. Case Studies in Thermal Engineering.
- Kakac, S., Liu, H., & Pramuanjaroenkij, A. (2012). Heat Exchangers: Selection, Rating, and Thermal Design. CRC Press.
- Shah, R.K., & Sekulic, D.P. (2003). Fundamentals of Heat Exchanger Design. Wiley.

Thank You
